מערכת ILT - מסמך תכנון מפורט והנחיות למימוש (Detailed Design - DDD v1.3 CODING-READY)

מהדורה סופית מוכנה לקידוד ע"י סוכני *AI (Gemini & Claude)*: ארכיטקטורת רכיבים, פרוטוקול בינארי, איתות *WebRTC*, סקריפטים ו-*Seed*, וחלוקת עבודה מודולרית

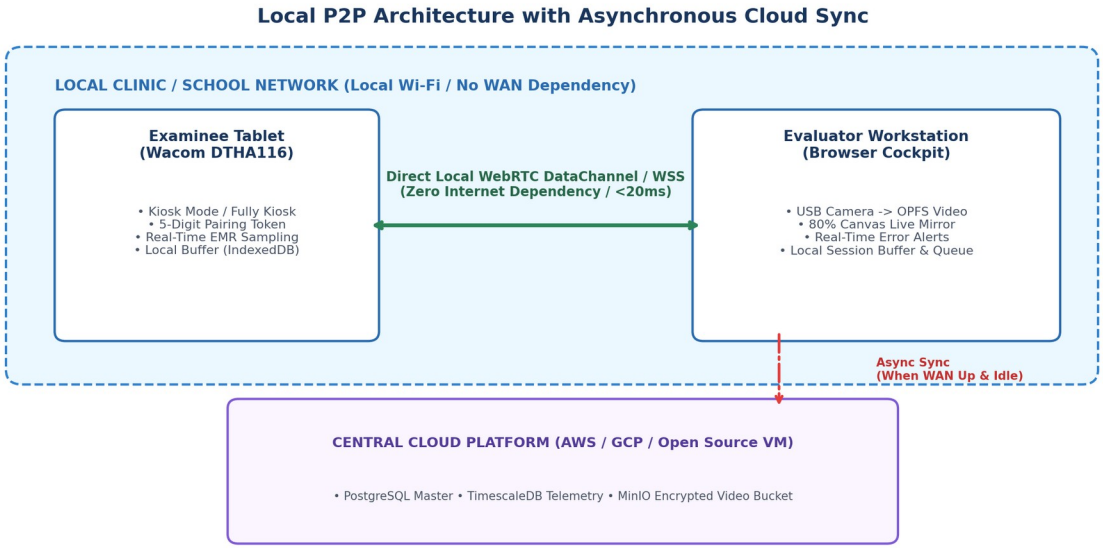
פרמטר מסמך	ערך והגדרה	פרמטר מסמך	ערך והגדרה
מזהה מסמך:	ILT-DDD-V1.3-CODING-READY-HEB	תאריך הפקה:	אוקטובר 2026 (מהדורה מוכנה לפיתוח)
בסיס פונקציונלי:	FRS v3.3 & Architecture Spec v5.3	מבצעי הקידוד:	Gemini & Claude (פיתוח אוטונומי ומודולרי)
סביבת בדיקות (Test):	מקומי Docker Compose + P2P LAN (אינו דורש Google Drive)	חומרה מיועדת:	Wacom MovinkPad 11 (טעינה רציפה נתמכת)
סטטוס מוכנות:	100% Coding-Ready (חוזים מפורשים ו-Seed מלא)	טיימרים:	ברירת מחדל 20 שנ' (שינוי באישור מנהל מחקר)

1. מטרת המסמך, סקירת המערכת ומטריצת טכנולוגיות

מסמך תכנון מפורט זה (DDD v1.3) הוכן כדי לאפשר לסוכני הפיתוח האוטונומיים (Gemini ו-Claude) להיכנס ישירות לקידוד המערכת ללא פערי מידע. המפרט כולל חוזי תקשורת בינאריים, סכמות JSON קשיחות, DDL מלא של מסד הנתונים ונתוני Seed התחלתיים, סביבת Docker Compose וסימולטור עט EMR לבדיקות.

2. ארכיטקטורת רכיבים, זרימת נתונים וערוצי WebRTC DataChannels

- ערוץ **WebRTC emr-stream** (לא-חוסם): ערוץ SCTP במצב בלתי-חוסם (ordered: false, maxRetransmits: 0) להעברת דגימות העט בקצב 133Hz בשיהוי נמוך מ-15ms.
- ערוץ **WebRTC control-events** (אמין): ערוץ SCTP במצב אמין (ordered: true) להעברת פקודות בקרה (OpCodes 0x01–0x1A), אירועי נעילה וטיימרים.



3. מבנה מאגר הקוד ותיקיות הפרויקט (**Monorepo Layout**)

```
/ilt-research-platform
├── /apps
│   ├── /student-tablet      אפליקציית הטאבלט לתלמיד (React + Canvas 2D + EMR Driver)
│   ├── /proctor-cockpit     עמדת המאבחן (Control Page, Live Cockpit, Comparative Dashboard)
│   └── /analyst-portal      פורטל אנליסט (Kinematic Replayer, Desk Video Sync, Superset Dashboards)
├── /services
│   ├── /gateway             שרת שער מקומי (Signaling, DB Cache, go2rtc Camera Proxy)
│   └── /analytics           שירות ניתוח קינמטי (cDTW לשורה  וייצוא DuckDB/Parquet (FastAPI)
├── /packages
│   ├── /shared-types        מודלי נתונים, ממשקים ו-Enums משותפים (TypeScript)
│   ├── /emr-protocol        מודול סריאליזציה בינארית (Byte Struct-) ומילון OpCodes
│   └── /template-schemas    סכמות JSON של תבניות כתיבה (גריד 3x5 מחברת שורות
אותיות ט')
└── /docker                  תצורות Docker לפיתוח מקומי (Postgres+Timescale, MinIO, go2rtc)
```

4. פרוטוקול התקשורת הבינארי, פקטות ואיתות **WebRTC**

- מבנה פקטה של 24 בתים לדגימת עט: Little-Endian IEEE 754 float32 לקואורדינטות ולחץ, uint32 לזמן, int16 לזוויות, ו-uint8 לדגלים:

```
// C / C++ & TypedArray 24-Byte Binary EMR Sample Struct (Little-Endian IEEE 754)
#pragma pack(push, 1)
struct EMRSampleRecord {
    uint32_t timestamp_offset_ms; // 4 Bytes: Milliseconds elapsed since block launch
    float    x_coord;             // 4 Bytes: Normalized X sub-pixel coordinate (0.0000 - 1.0000)
    float    y_coord;             // 4 Bytes: Normalized Y sub-pixel coordinate (0.0000 - 1.0000)
    float    pressure;            // 4 Bytes: Normalized pen tip pressure (0.0000 - 1.0000)
    int16_t  tilt_altitude;       // 2 Bytes: Pen vertical angle relative to tablet plane (0 - 90 deg)
    int16_t  tilt_azimuth;        // 2 Bytes: Pen rotational angle around surface normal (0 - 360 deg)
    uint8_t  stroke_flags;        // 1 Byte: Bit 0: PenDown, Bit 1: InAirHover, Bit 2: Eraser, Bit 3: TargetHit
    uint8_t  active_house_id;     // 1 Byte: Identifier of active house/cell (1 to 15)
    uint16_t reserved_padding;    // 2 Bytes: Alignment padding (0x0000)
};
#pragma pack(pop)
```

- חוזה איתות **WebRTC Signaling (JSON)**: פרוטוקול איתות מבוסס WebSockets לפתיחת ערוץ P2P בין הטאבלט לקוקפיט:

```
// WebRTC Signaling Protocol Contracts (WebSocket / JSON)
// Tablet -> Gateway (Signaling): Join session room.
{
  "event": "JOIN_ROOM",
  "payload": {
    "role": "TABLET",
    "pin": "84920",
    "device_model": "Wacom MovinkPad 11 / DTHA116"
  }
}

// Gateway -> Proctor Cockpit: Notify tablet joined.
{
  "event": "PEER_JOINED",
  "payload": {
    "peer_id": "tablet_client_uuid",
    "role": "TABLET"
  }
}

// Proctor Cockpit -> Gateway -> Tablet: WebRTC SDP Offer.
{
  "event": "WEBRTC_OFFER",
  "payload": {
    "...sdp": "v=0\r\no=- 420 2 IN IP4 127.0.0.1",
    "type": "offer"
  }
}

// Tablet -> Gateway -> Cockpit: WebRTC SDP Answer.
{
  "event": "WEBRTC_ANSWER",
  "payload": {
    "...sdp": "v=0\r\no=- 421 2 IN IP4 127.0.0.1",
    "type": "answer"
  }
}

// Bidirectional: ICE Candidate Exchange.
{
  "event": "ICE_CANDIDATE",
  "payload":
```

```
","...candidate": "candidate:842163049 1 udp 1677729535 192.168.1.105 52834 typ host"
                                                                ","sdpMid": "0"
                                                              sdpMLineIndex": 0"
                                                                        {
                                                                         {
```

5. מנוע הקנבס, גיאומטריית תבניות ומנגנון הנעילה

```
                                                                         }
                                                  ","   schema": "http://json-schema.org/draft-07/schema$"
                                                             ","title": "ILTDynamicTemplateDefinition"
                                                                  ","type": "object"
                                                                  } :"properties"
                                                        template_id": { "type": "string" }"
                                                      template_name": { "type": "string" }"
                                  ruling_type": { "type": "string", "enum": ["NOTEBOOK_10_ROWS", "NOTEBOOK_14_ROWS","
                                                  "NOTEBOOK_1_ROW", "GRID_3X5", "FREE_CANVAS"] }
                                  target_grade_level": { "type": "string", "default": "ALL_GRADES" }"
                                                                  } :"canvas_schema"
                                                                 ","type": "object"
                                                                   } :"properties"
                                                          num_lines": { "type": "integer" }"
                                                       line_spacing_px": { "type": "number" }"
                                          margin_side": { "type": "string", "enum": ["RIGHT", "LEFT", "NONE"] }"
                                                       margin_offset_px": { "type": "number" }"
                                  active_row_indices": { "type": "array", "items": { "type": "integer" } }"
                                                                                          ,{
                                   required": ["num_lines", "line_spacing_px", "margin_side", "active_row_indices"]"
                                                                                         ,{
                                                                  } :"rules_schema"
                                                                 ","type": "object"
                                                                   } :"properties"
                                              max_internal_pen_lifts": { "type": "integer", "default": 0 }"
                                                 width_nominal_mm": { "type": "number", "default": 15.0 }"
                                                   width_tolerance_mm": { "type": "number", "default": 2.0 }"
                                                forward_only_locking": { "type": "boolean", "default": true }"
                                                                                        {
                                                                                         {
                                                                                         ,{
                                 required": ["template_id", "template_name", "ruling_type", "canvas_schema", "rules_schema"]"
                                                                                         {
```

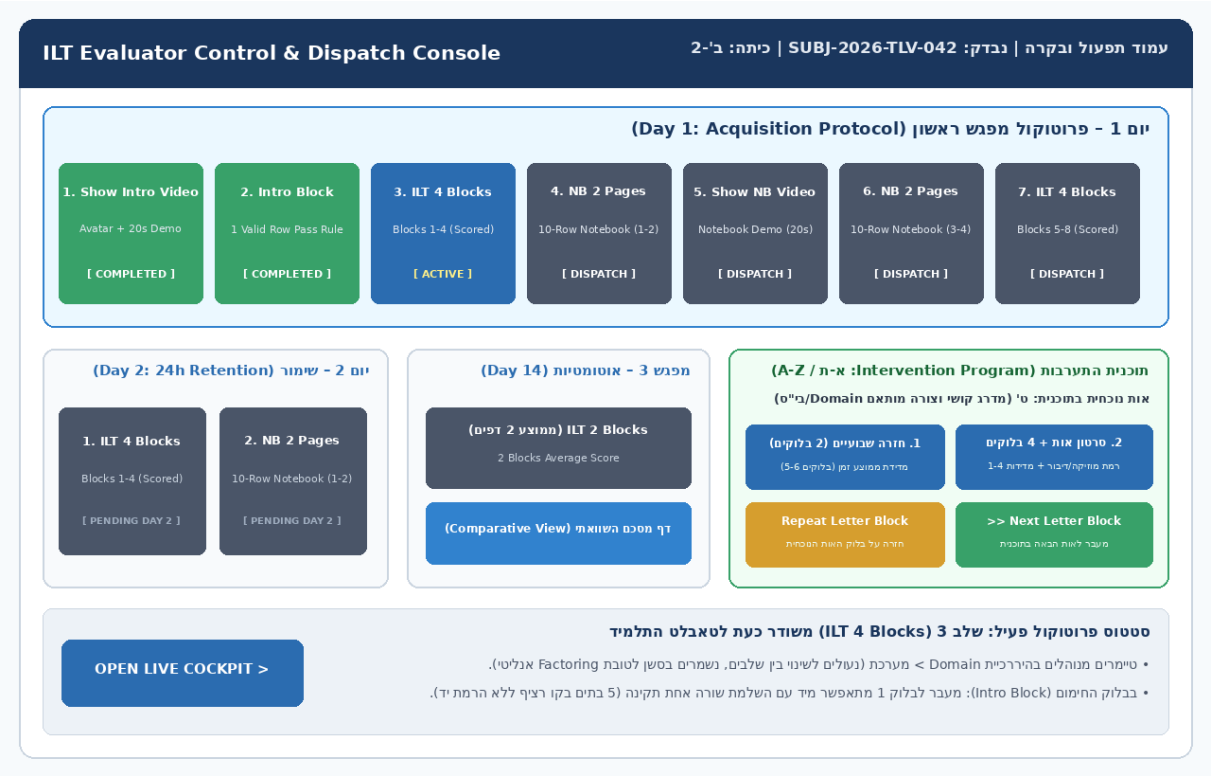


איור 2: קנבס הטאבלט לתלמיד (נעילה קדימה ורציפות בטעינה)

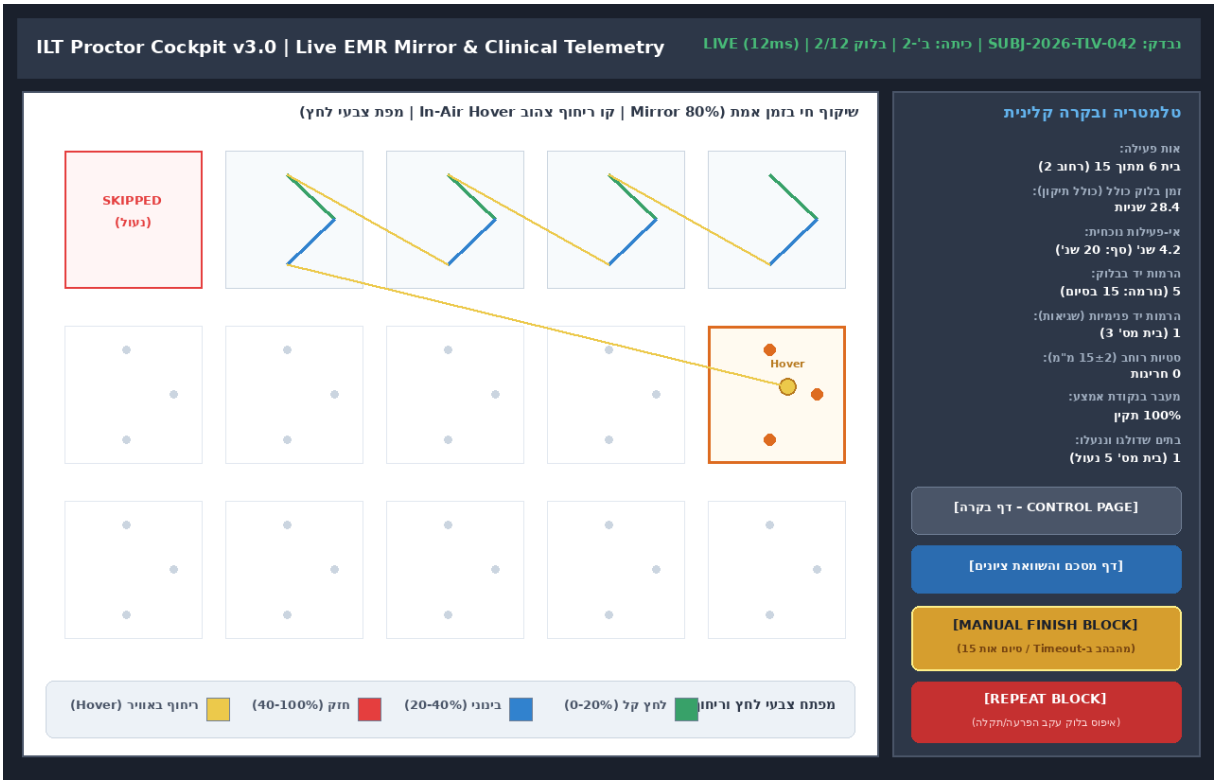


איור 3: תבנית מחברת שורות (*NB2Pages*) – מותאמת באופן מלא לאיור 13 מהמחקר המקורי

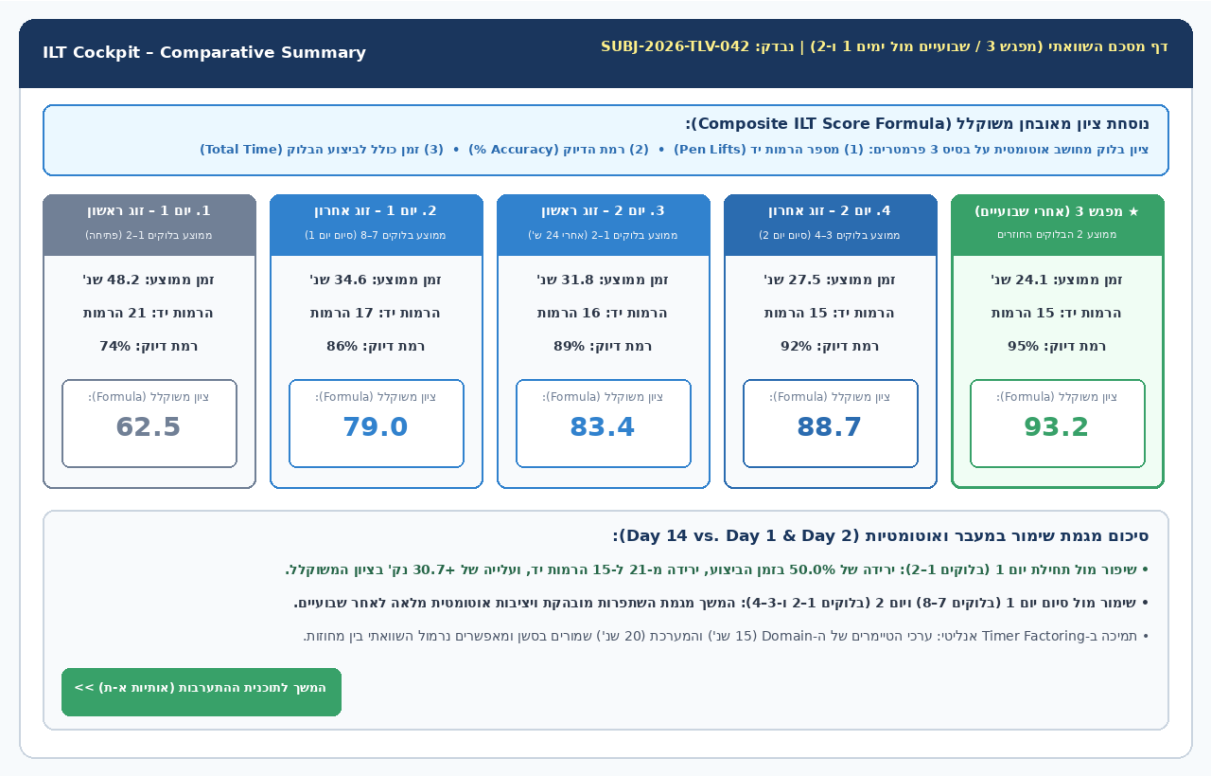
6. מכונות מצבים, קוקפיט מאבחן ודף מסכם



איור 4: עמוד תפעול ובקרה (*Control Page*)



איור 5: קוקפיט המאבחן 1:1



איור 6: דף מסכם השוואתי המציג 3 מדדים נפרדים (הנחיית פרופ' אסתי עדי-יפה)

7. סכמת מסד נתונים מלאה ונתוני **Seed** התחלתיים

סכמת ה-DDL המלאה של PostgreSQL 16 + TimescaleDB:

```
--
================================================================================
ILT PRODUCTION DDL (PostgreSQL 16 + TimescaleDB) - Release 5.3 / DDD v1.2 --
--
================================================================================
;"CREATE EXTENSION IF NOT EXISTS "uuid-ossp
;"CREATE EXTENSION IF NOT EXISTS "timescaledb

System Configuration & Global Defaults .    --
 CREATE TABLE system_config
config_id INT PRIMARY KEY DEFAULT 1
    in_transition_delay_sec INT NOT NULL DEFAULT 20, -- Default In-Transition: 20s
    inactivity_timeout_sec INT NOT NULL DEFAULT 20,  -- Default Inactivity Timeout: 20s
    post_block_timeout_sec INT NOT NULL DEFAULT 30,  -- Timeout post-15th letter: 30s
notebook_page_timeout_sec INT NOT NULL DEFAULT 60
video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE
require_research_director_approval BOOLEAN NOT NULL DEFAULT TRUE
score_weight_pen_lifts_pct NUMERIC(5,2) NOT NULL DEFAULT 35.00
score_weight_accuracy_pct NUMERIC(5,2) NOT NULL DEFAULT 40.00
```

```
    score_weight_time_pct NUMERIC(5,2) NOT NULL DEFAULT 25.00
    CONSTRAINT chk_weights_sum_100 CHECK
= (score_weight_pen_lifts_pct + score_weight_accuracy_pct + score_weight_time_pct
                                                                    .
                                                                  ,(
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
                                                                 ;(

Domains Table (with Research Director Authorization for Overrides) .   --
                                                    CREATE TABLE domains
                                                  domain_id SERIAL PRIMARY KEY
                                               domain_name VARCHAR(100) NOT NULL UNIQUE
                                             override_in_transition_sec INT NULL
                                          override_inactivity_timeout_sec INT NULL
                                             override_video_enabled BOOLEAN NULL
                                  timer_approved_by_research_director VARCHAR(100) NULL
                                  timer_approval_timestamp TIMESTAMP WITH TIME ZONE NULL
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
                                                                 ;(

                                                Dynamic Templates Engine .   --
                                                  CREATE TABLE dynamic_templates
                                               template_id VARCHAR(50) PRIMARY KEY
                                             template_name VARCHAR(100) NOT NULL
                                          template_category VARCHAR(30) NOT NULL
                                               ruling_type VARCHAR(30) NOT NULL
                                  ,'target_grade_level VARCHAR(30) DEFAULT 'ALL_GRADES
                                               canvas_schema JSONB NOT NULL
                                                rules_schema JSONB NOT NULL
                                              is_active BOOLEAN DEFAULT TRUE
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
                                                                 ;(

                            Canonical Models for Freehand Row 3 Evaluation .   --
                                                    CREATE TABLE letter_canonical_models
                                                 letter_id VARCHAR(10) PRIMARY KEY
                                               language_code VARCHAR(5) NOT NULL
                                                letter_char VARCHAR(5) NOT NULL
                                        expected_strokes_count INT NOT NULL DEFAULT 1
                                        canonical_aspect_ratio NUMERIC(5,3) NOT NULL
                                               canonical_points JSONB NOT NULL
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
```

```
);

-- Sessions Table (3 Metrics Preserved Independently + Timer Snapshots).
CREATE TABLE sessions
(),session_id UUID PRIMARY KEY DEFAULT gen_random_uuid
session_number BIGSERIAL UNIQUE,
domain_id INT NOT NULL REFERENCES domains(domain_id),
school_name VARCHAR(150) NOT NULL,
grade_class VARCHAR(50) NOT NULL,
evaluator_name VARCHAR(120) NOT NULL,
assessment_date DATE NOT NULL DEFAULT CURRENT_DATE,
examinee_alias VARCHAR(50) NOT NULL,
examinee_initials VARCHAR(5) NOT NULL,
'student_language VARCHAR(5) DEFAULT 'he,
session_stage VARCHAR(30) NOT NULL,
is_charging_during_test BOOLEAN DEFAULT FALSE,
sys_in_transition_sec INT NOT NULL DEFAULT 20,
dom_in_transition_sec INT NOT NULL DEFAULT 20,
sys_inactivity_sec INT NOT NULL DEFAULT 20,
dom_inactivity_sec INT NOT NULL DEFAULT 20,
video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE,
warmup_row_passed BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- Assessment Blocks Table (Preserving 3 Distinct Metrics Independently).
CREATE TABLE session_blocks
(
block_id BIGSERIAL PRIMARY KEY,
session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
block_type VARCHAR(30) NOT NULL,
block_index INT NOT NULL,
target_letter VARCHAR(5) NULL,
pen_lifts_count INT NOT NULL,
accuracy_pct NUMERIC(5,2) NOT NULL,
total_duration_ms INT NOT NULL,
composite_score NUMERIC(6,2) NULL,
is_repeated_disrupted BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- High-Frequency EMR Telemetry (TimescaleDB Hypertable with 1-Day Chunks).
CREATE TABLE pen_telemetry_chunks
```

```
    recorded_at TIMESTAMP WITH TIME ZONE NOT NULL
    session_id UUID NOT NULL
    block_index INT NOT NULL
    house_id INT NOT NULL
    time_offset_ms INT NOT NULL
    x NUMERIC(8,4) NOT NULL
    y NUMERIC(8,4) NOT NULL
    pressure NUMERIC(6,5) NOT NULL
    tilt_altitude SMALLINT NOT NULL
    tilt_azimuth SMALLINT NOT NULL
    is_hover BOOLEAN NOT NULL
    is_pen_down BOOLEAN NOT NULL
);
SELECT create_hypertable('pen_telemetry_chunks', 'recorded_at', chunk_time_interval => INTERVAL
'1 day');
ALTER TABLE pen_telemetry_chunks SET (timescaledb.compress, timescaledb.compress_segmentby =
'session_id, block_index');
SELECT add_compression_policy('pen_telemetry_chunks', INTERVAL '7 days');

-- Video Replay Records (Desk & Hands RTSP / MP4 Ingestion).
CREATE TABLE session_video_records (
    video_id BIGSERIAL PRIMARY KEY,
    session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
    camera_model VARCHAR(100) DEFAULT 'Wireless IP Camera (RTSP)',
    storage_bucket VARCHAR(50) DEFAULT 'ilt-session-videos',
    video_object_key TEXT NOT NULL,
    duration_sec NUMERIC(8,2) NOT NULL,
    start_time_utc TIMESTAMP WITH TIME ZONE NOT NULL,
    fps NUMERIC(4,2) DEFAULT 25.00,
    resolution VARCHAR(20) DEFAULT '1280x720',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- Analytical Cohort Slices View (for Apache Superset, Metabase, DuckDB, Jupyter).
CREATE OR REPLACE VIEW v_analyst_cohort_slices AS
SELECT
    s.session_id,
    s.session_number,
    s.examinee_alias,
    s.examinee_initials,
    d.domain_name,
    s.school_name,
    s.grade_class,
```

```
    s.dominant_hand
    s.student_language
    s.session_stage
    s.assessment_date
    s.sys_in_transition_sec
    s.dom_in_transition_sec
    AS (s.dom_in_transition_sec::numeric / s.sys_in_transition_sec::numeric
    timer_factoring_ratio
    sb.block_id
    sb.block_type
    sb.block_index
    sb.target_letter
    sb.pen_lifts_count
    sb.accuracy_pct
    sb.total_duration_ms
    AS total_duration_sec (sb.total_duration_ms::numeric / 1000.0
    (sb.total_duration_ms::numeric / 1000.0
s.sys_in_transition_sec::numeric /)
    AS factored_duration_sec ((s.dom_in_transition_sec::numeric
    sb.composite_score
    sb.is_repeated_disrupted
FROM sessions s
JOIN domains d ON s.domain_id = d.domain_id
JOIN session_blocks sb ON s.session_id = sb.session_id;
```

סקריפט נתוני Seed ראשוניים (הגדרות טיימרים 20 שנ', מחוזות ותבניות מובנות):

```
--
============================================================
============================
-- SEED DATA & INITIAL CONFIGURATION (PostgreSQL 16)
--
============================================================
============================

-- Insert System Default Configuration (Timers: 20s, Sum(weights) = 100%).
INSERT INTO system_config (
    config_id, in_transition_delay_sec, inactivity_timeout_sec, post_block_timeout_sec,
    notebook_page_timeout_sec, video_recording_enabled, require_research_director_approval,
    score_weight_pen_lifts_pct, score_weight_accuracy_pct, score_weight_time_pct
) VALUES (
    1, TRUE, TRUE, 35.00, 40.00, 25.00
) ON CONFLICT (config_id) DO NOTHING;

-- Insert Default Districts (Domains).
INSERT INTO domains (domain_name, override_in_transition_sec, override_inactivity_timeout_sec)
```

```
VALUES
    ('Central District', 'מחוז מרכז', NULL, NULL),
    ('Tel Aviv District', 'מחוז תל אביב', NULL, NULL),
    ('Haifa District', 'מחוז חיפה', NULL, NULL),
    ('Jerusalem District', 'מחוז ירושלים', NULL, NULL),
    ('Northern District', 'מחוז צפון', NULL, NULL),
    ('Southern District', 'מחוז דרום', NULL, NULL)
ON CONFLICT (domain_name) DO NOTHING;

-- Insert Standard Dynamic Templates.
INSERT INTO dynamic_templates (
    template_id, template_name, template_category, ruling_type, target_grade_level,
    canvas_schema, rules_schema
) VALUES
(
    'TEMPLATE_ILT_GRID_3X5',
    'Standard ILT 3x5 Grid (רשת בתים לאות מומצאת)',
    'GRID_ILT',
    'GRID_3X5',
    'ALL_GRADES',
    '{"rows": 3, "cols": 5, "box_w_px": 170, "box_h_px": 145, "dot_hit_radius_mm": 3.5}',
    '{"max_internal_pen_lifts": 0, "width_nominal_mm": 15.0, "width_tolerance_mm": 2.0, "forward_only_locking": true}'
),
(
    'TEMPLATE_NOTEBOOK_10_ROWS',
    'Authentic 10-Row Israeli Elementary Notebook (מחברת שורות לכיתות יסודי)',
    'NOTEBOOK_LINED',
    'NOTEBOOK_10_ROWS',
    'ALL_GRADES',
    '{"num_lines": 16, "line_spacing_px": 35.0, "margin_side": "RIGHT", "active_row_indices": [4, 7, 10]}',
    '{"max_internal_pen_lifts": 0, "time_measurement_only": true, "letters_per_row": 5}'
),
(
    'TEMPLATE_INTERVENTION_LETTER_TET',
    'Intervention Scaffolding Template (האות העברית ט'')',
    'INTERVENTION_LETTER',
    'GRID_3X5',
    'ALL_GRADES',
    '{"row1": "MODEL_PLUS_TARGET_DOTS", "row2": "START_DOT_ONLY", "row3": "BLANK_FREEHAND"}',
    '{"max_internal_pen_lifts": 0, "dtw_accuracy_evaluation_row3": true, "nominal_aspect_ratio":
```

```
'{0.85
(
;ON CONFLICT (template_id) DO NOTHING
```

8. מימוש מנוע שחזור קינמטי ושאילתות חיתוך (**Analyst Replay & Cohort Engine**)

- קוד **TypeScript** לנגן השחזור הקינמטי (**KinematicSessionReplayer**): מימוש שחזור משיכות ב- Canvas 2D, בקרת מהירות (0.25x-4x), וסנכרון וידאו שולחן/ידיים:

```
// Kinematic Session Replay Engine (TypeScript / React Canvas 2D)
export class KinematicSessionReplayer {
    private currentFrameIndex = 0;
    private playbackSpeed = 1.0;
    private isPlaying = false;
    private animFrameId: number | null = null;

    constructor(
        private canvas: HTMLCanvasElement,
        private samples: any[],
        private videoElement?: HTMLVideoElement,
        private onProgress?: (timeMs: number, sample: any) => void
    ) {}

    public play(): void {
        this.isPlaying = true;
        if (this.videoElement) this.videoElement.play();
        this.renderLoop();
    }

    public pause(): void {
        this.isPlaying = false;
        if (this.animFrameId) cancelAnimationFrame(this.animFrameId);
        if (this.videoElement) this.videoElement.pause();
    }

    public setSpeed(speed: number): void {
        this.playbackSpeed = speed;
        if (this.videoElement) this.videoElement.playbackRate = speed;
    }

    public seek(targetTimeMs: number): void {
        this.currentFrameIndex = this.samples.findIndex(s => s.timestamp_offset_ms >=
            targetTimeMs);
        if (this.currentFrameIndex === -1) this.currentFrameIndex = this.samples.length - 1;
```

```
        ;if (this.videoElement) this.videoElement.currentTime = targetTimeMs / 1000.0
                                                        ;()this.renderUpToCurrentFrame
                                                                                    {


                                                    } <= private renderLoop = (): void
                                                            ;if (!this.isPlaying) return
                                                ;const currentSample = this.samples[this.currentFrameIndex]
                                                            ;this.drawStrokeSegment(currentSample)
;if (this.onProgress) this.onProgress(currentSample.timestamp_offset_ms, currentSample)

                                                ;this.currentFrameIndex += Math.max(1, Math.round(this.playbackSpeed))
                                                } if (this.currentFrameIndex < this.samples.length)
                                                ;this.animFrameId = requestAnimationFrame(this.renderLoop)
                                                                                    } else {
                                                                                ;()this.pause
                                                                                        {
                                                                                         ;{


                                                                } private drawStrokeSegment(s: any): void
                                                                ;!const ctx = this.canvas.getContext('2d')
                                                                  ;const x = s.x_coord * this.canvas.width
                                                                 ;const y = s.y_coord * this.canvas.height
                                                                  if (s.stroke_flags & 0x01) { // Pen Down
                                                ;ctx.fillStyle = this.getPressureColor(s.pressure)
                                                                                ;()ctx.beginPath
                                                ;ctx.arc(x, y, 2 + s.pressure * 4, 0, Math.PI * 2)
                                                                                    ;()ctx.fill
                                                 else if (s.stroke_flags & 0x02) { // In-Air Hover {
                                                   ctx.fillStyle = '#ECC94B'; // Yellow hover line
                                                                ;ctx.fillRect(x - 1, y - 1, 2, 2)
                                                                                        {
                                                                                          {


                                                                } private getPressureColor(p: number): string
                                                            if (p < 0.20) return '#38A169'; // Light Green
                                                            if (p < 0.40) return '#3182CE'; // Medium Blue
                                                               return '#E53E3E';               // Firm Red
                                                                                        {
                                                                                      {
```

- קוד **Python** לייצוא חיתוכי נתונים ל-**Parquet** דרך **DuckDB**: שאילתות חיתוך מהירות במיוחד וייצוא קבצי Parquet ישירות לחוקרים:

```
Python Cohort Analytics & Parquet / DuckDB Export Engine (FastAPI)
                                                                import duckdb
```

```
import pandas as pd
from fastapi import APIRouter, Response

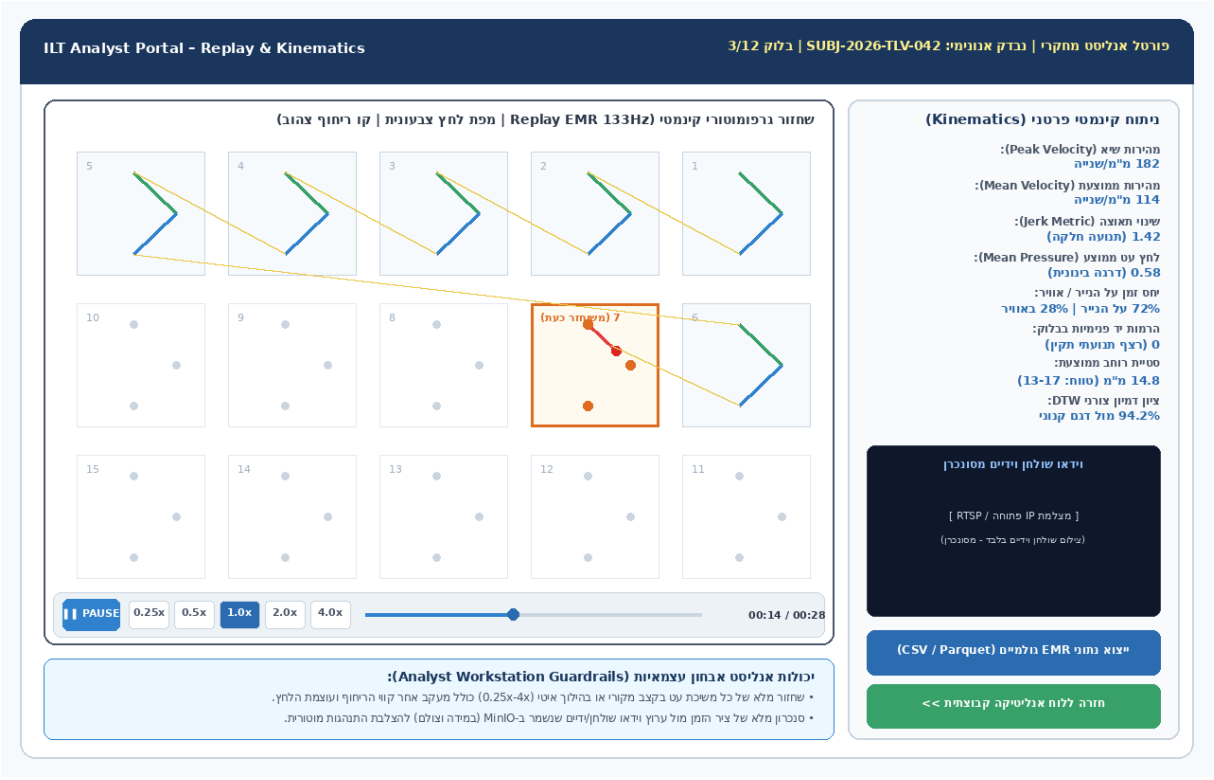
router = APIRouter(prefix="/api/v1/analyst")

@router.get("/export/parquet")
def export_cohort_parquet(domain: str = None, grade: str = None, ses_low: bool = None):
    con = duckdb.connect()
    con.execute("INSTALL postgres; LOAD postgres;")
    con.execute("ATTACH 'dbname=ilt_research user=ilt_admin host=db' AS pg (TYPE POSTGRES);")

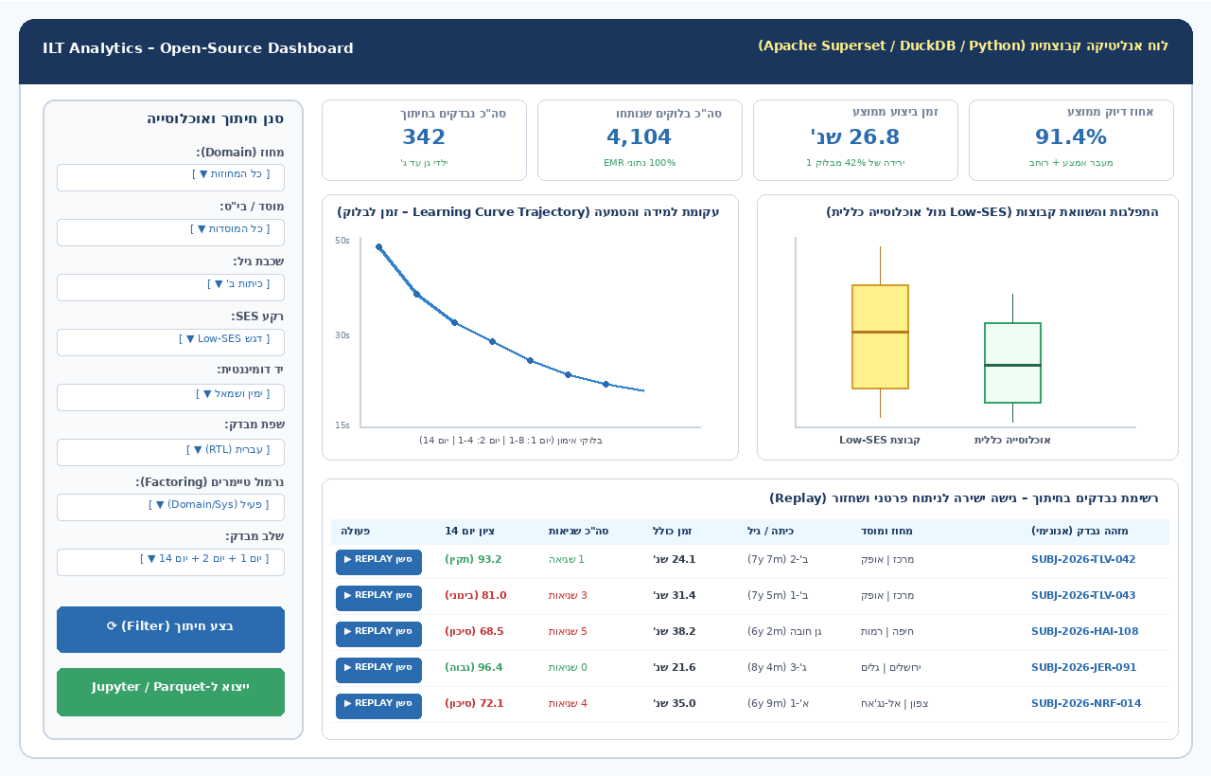
    query = "SELECT * FROM pg.v_analyst_cohort_slices WHERE 1=1"
    if domain: query += f" AND domain_name = '{domain}'"
    if grade: query += f" AND grade_class LIKE '{grade}%'"

    df = con.execute(query).df()
    parquet_bytes = df.to_parquet(index=False)

    return Response(
        content=parquet_bytes,
        media_type="application/octet-stream",
        headers={"Content-Disposition": "attachment; filename=ilt_cohort_data.parquet"}
    )
```



איור 7: פורטל אנליסט מחקרי – נגן שחזור קינמטי פרטני (*Replay Engine*) מסונכרן לוידאו



איור 8: לוח אנליטיקה קבוצתית וחיתוכים מרובי-משתנים (*Apache Superset / DuckDB / Jupyter*)

9. אלגוריתמיקה מפורטת ותוכנית ההתערבות

```
import numpy as np
from scipy.spatial import procrustes
from fastdtw import fastdtw

def evaluate_freehand_row3(stroke_points: np.ndarray, canonical_points: np.ndarray,
                           canonical_aspect_ratio: float) -> dict:
    if len(stroke_points) < 10:
        return {"composite_accuracy": 0.0, "diagnostic_tag": "EMPTY_OR_NOISE", "valid": False}

    diffs = np.diff(stroke_points, axis=0)
    dists = np.hypot(diffs[:, 0], diffs[:, 1])
    cum_dist = np.insert(np.cumsum(dists), 0, 0)
    total_len = cum_dist[-1]
    even_spaces = np.linspace(0, total_len, 100)
    resampled_stroke = np.column_stack(
        (np.interp(even_spaces, cum_dist, stroke_points[:, 0]),
         np.interp(even_spaces, cum_dist, stroke_points[:, 1]))
    )

    norm_stroke = (resampled_stroke - np.mean(resampled_stroke, axis=0)) /
                  (np.std(resampled_stroke) + 1e-6)
    norm_canonical = (canonical_points - np.mean(canonical_points, axis=0)) /
                     (np.std(canonical_points) + 1e-6)
```

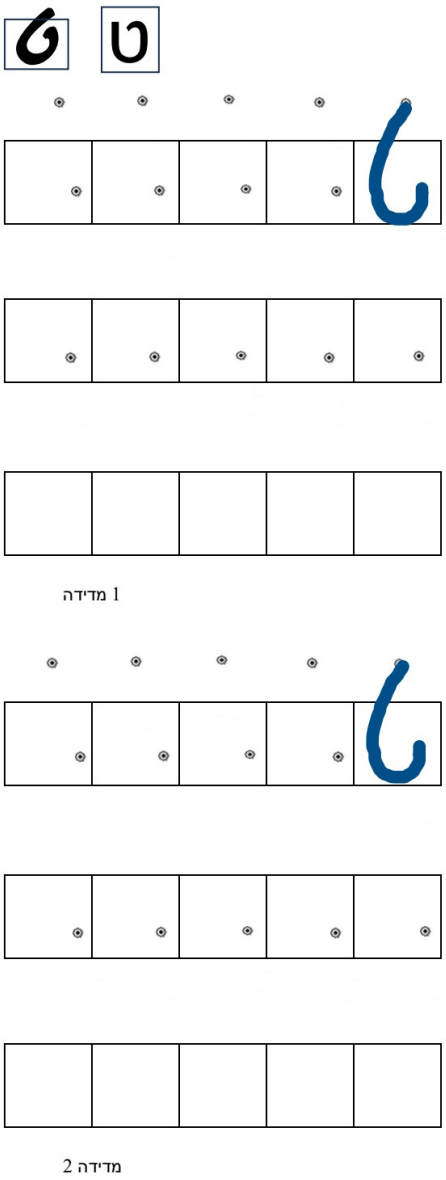
```
                disparity = procrustes(norm_canonical, norm_stroke) ,_ ,_
        procrustes_score = max(0.0, min(100.0, 100.0 * (1.0 - disparity * 2.2)))

                dtw_dist, _ = fastdtw(norm_stroke, norm_canonical, dist=2)
        dtw_score = max(0.0, min(100.0, 100.0 * (1.0 - (dtw_dist / 100.0))))

              w, h = np.ptp(stroke_points[:, 0]), np.ptp(stroke_points[:, 1])
                                        actual_ratio = w / (h + 1e-6)
ratio_dev_pct = abs(actual_ratio - canonical_aspect_ratio) / canonical_aspect_ratio * 100.0

       composite_acc = 0.55 * procrustes_score + 0.30 * dtw_score + 0.15 * max(0.0, 100.0 -
                                                                    ratio_dev_pct)
 tag = "ACCURATE" if composite_acc >= 80.0 else ("PROPORTION_DISTORTION" if ratio_dev_pct >
                                                    25.0 else "SHAPE_DEFORMATION")

                                                          } return
                                    composite_accuracy": round(composite_acc, 2)"
                                    procrustes_score": round(procrustes_score, 2)"
                                        dtw_distance": round(dtw_dist, 2)"
                                aspect_ratio_deviation_pct": round(ratio_dev_pct, 2)"
                                        diagnostic_tag": tag"
                                                valid": True"
                                                        {
```



איור 9: דוגמת דף אימון של תוכנית ההתערבות (האות ט')

10. נספח הנחיות למתכנת (**Developer Onboarding & AI Implementation Guide**)

- סביבת **Docker Compose** מקומית: הרמת TimescaleDB, MinIO ו-go2rtc בפקודה אחת:

```
version: '3.8'
services:
  db:
    image: timescale/timescaledb:latest-pg16
    container_name: ilt_db
    environment:
      POSTGRES_DB: ilt_research
      POSTGRES_USER: ilt_admin
      POSTGRES_PASSWORD: ilt_secret_password
    ports:
      - "          :          "
    volumes:
```

```
      - pgdata:/var/lib/postgresql/data
    restart: always

  minio:
    image: minio/minio:latest
    container_name: ilt_minio
    command: server /data --console-address ":9001"
    environment:
      MINIO_ROOT_USER: minio_admin
      MINIO_ROOT_PASSWORD: minio_secret_password
    ports:
      - "          :          "
      - "          :          "
    volumes:
      - miniodata:/data
    restart: always

  go2rtc:
    image: alexxit/go2rtc:latest
    container_name: ilt_cms_gateway
    ports:
      - "          :          "    Management & WebRTC API
      - "          :          "    RTSP Relay Port
      - "          :          "    WebRTC Live Port
    volumes:
      - ./go2rtc.yaml:/config/go2rtc.yaml
    restart: always

volumes:
  pgdata:
  miniodata:
```

- סימולטור עט **EMR** לפיתוח בדפדפן שולחני: מיפוי אירועי עכבר לדגימות EMR מדומות עם לחץ וריחוף:

```
// Mock Wacom EMR Stylus Simulator for Desktop Browser Development
export class MockEMRStylusSimulator {
  private isMouseDown = false;
  private startTime = 0;

  constructor(private canvas: HTMLCanvasElement, private onSample: (s: any) => void) {
    this.canvas.addEventListener('mousedown', (e) => {
      this.isMouseDown = true;
      this.startTime = performance.now();
      this.emit(e, 0.45, true, false);
```

```
                                                                    ;({
                                                  } <= window.addEventListener('mouseup', (e)
                                                                  } if (this.isMouseDown)
                                                              ;this.isMouseDown = false
                                                      ;this.emit(e, 0.0, false, false)
                                                                                    {
                                                                    ;({
                                                } <= this.canvas.addEventListener('mousemove', (e)
const pressure = this.isMouseDown ? 0.55 + 0.15 * Math.sin(performance.now() / 80) :
                                                                                  ;0.0
                                                          ;const isHover = !this.isMouseDown
                                                ;this.emit(e, pressure, this.isMouseDown, isHover)
                                                                    ;({
                                                                        {


    } private emit(e: MouseEvent, pressure: number, isPenDown: boolean, isHover: boolean)
                                          ;()const rect = this.canvas.getBoundingClientRect
                                            ;const x = (e.clientX - rect.left) / rect.width
                                            ;const y = (e.clientY - rect.top) / rect.height
                                                                  })this.onSample
                                  time_offset_ms: Math.round(performance.now() - this.startTime)
                                                                  x: Number(x.toFixed(5))
                                                                  y: Number(y.toFixed(5))
                                                pressure: Number(pressure.toFixed(4))
                                                                  tilt_altitude: 60
                                                                  tilt_azimuth: 180
                                                                is_pen_down: isPenDown
                                                                      is_hover: isHover
                                                                    ;({
                                                                        {
                                                                          {
```